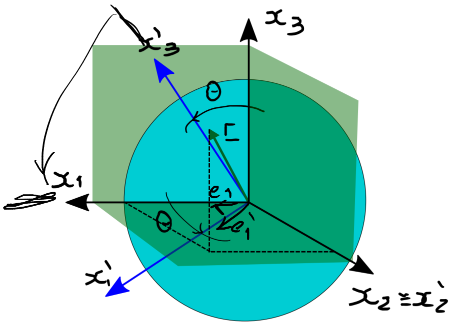
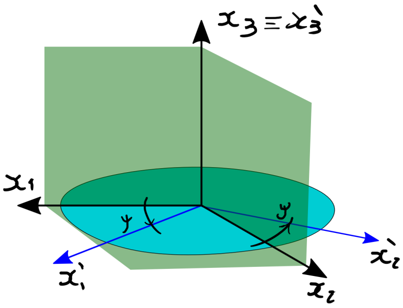
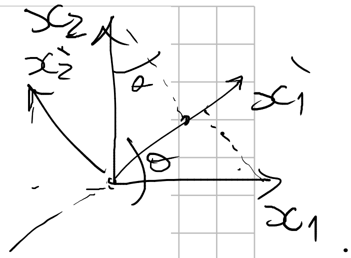


Rotación alrededor del Eje y



Rotación alrededor del Eje z



$x'_1 = \cos\theta . x_1 + \underbrace{\cos(\pi/2 - \theta)}_{\text{sen}\,\theta} . x_2$

$$\boldsymbol{\beta_1} = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix} \qquad \boldsymbol{\beta_2} = \begin{bmatrix} \cos\psi & \sin\psi & 0 \\ -\sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} \cos\ \text{sen}\theta \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$\underbrace{\underline{\underline{B}}_1 \cdot \underline{\underline{B}}_1^T = \underline{\underline{I}}}_{\text{Verifica}}$

$(\underline{\underline{B}}_1)_{ij} = \underline{e}'_i \cdot \underline{e}_j = \cos\theta$

$B_{i2} = \underline{e}'_i \cdot \underline{e}_2 = 0 \quad \text{si } i \neq 2$

$B_{1\,22} = \cos(0) = 1$

$\det(\underline{\underline{B}}) = 1$

$B_{1\,11} = e'_1\, e_1 = \cos(e'_1, e_2) = \cos\theta$

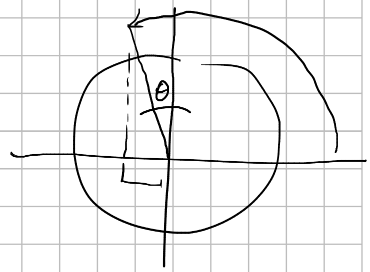
$B_{1\,13} = \cos(\theta + \frac{\pi}{2}) = -\text{sen}\,\theta$

$B_{1\,2i} = e'_2 \cdot e_i = 0 \quad \text{si } i \neq 2$

$e'_3 \cdot e_1$

$B_{31} = \cos(\frac{\pi}{2} - \theta) = \text{sen}\,\theta$

$B_{33} = \cos\theta$



- Primera transformación: $\boldsymbol{r} \xrightarrow{\beta_1} \boldsymbol{r}' \xrightarrow{\beta_2} \boldsymbol{r}''$

$$\left.\begin{aligned} \boldsymbol{r}' &= \boldsymbol{\beta_1} \cdot \boldsymbol{r} \\ \boldsymbol{r}'' &= \boldsymbol{\beta_2} \cdot \boldsymbol{r}' \end{aligned}\right\} \implies \boldsymbol{r}'' = \boldsymbol{\beta_2} \cdot \boldsymbol{\beta_1} \cdot \boldsymbol{r}$$

- Segunda transformación: $\boldsymbol{r} \xrightarrow{\beta_2} \boldsymbol{r}''' \xrightarrow{\beta_1} \boldsymbol{r}^{IV}$

$$\left.\begin{aligned} \boldsymbol{r}''' &= \boldsymbol{\beta_2} \cdot \boldsymbol{r} \\ \boldsymbol{r}^{IV} &= \boldsymbol{\beta_1} \cdot \boldsymbol{r}''' \end{aligned}\right\} \implies \boldsymbol{r}^{IV} = \boldsymbol{\beta_1} \cdot \boldsymbol{\beta_2} \cdot \boldsymbol{r}$$

Para que esta rotación sea conmutativa debería cumplirse que $\boldsymbol{r}'' = \boldsymbol{r}^{IV} \quad \forall \boldsymbol{r}$:

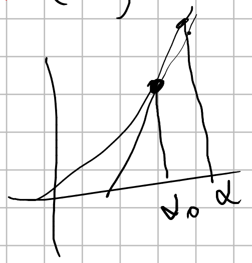
$$\boldsymbol{r}'' = \boldsymbol{r}^{IV} \quad \forall \boldsymbol{r} \implies \boldsymbol{\beta_2} \cdot \boldsymbol{\beta_1} \cdot \boldsymbol{r} = \boldsymbol{\beta_1} \cdot \boldsymbol{\beta_2} \cdot \boldsymbol{r} \quad \forall \boldsymbol{r} \implies \boldsymbol{\beta_2} \cdot \boldsymbol{\beta_1} = \boldsymbol{\beta_1} \cdot \boldsymbol{\beta_2}$$

Por lo que se realiza las multiplicación matriciales:

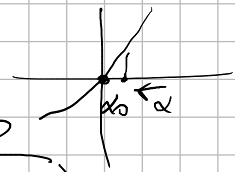
$$\boldsymbol{\beta}_2 \cdot \boldsymbol{\beta}_1 = \begin{bmatrix} \cos\theta\ \cos\psi & \sin\psi & \sin\theta\ \cos\psi \\ -\cos\theta\ \sin\psi & \cos\theta & -\sin\theta\ \sin\psi \\ -\sin\theta & 0 & \cos\theta \end{bmatrix} \qquad \boldsymbol{\beta}_1 \cdot \boldsymbol{\beta}_2 = \begin{bmatrix} \cos\theta\ \cos\psi & \cos\theta\ \sin\psi & \sin\theta \\ -\sin\psi & \cos\psi & 0 \\ -\sin\theta\ \cos\psi & -\sin\theta\ \sin\psi & \cos\theta \end{bmatrix}$$

$\alpha \rightarrow$ ángulo genérico)

$$\cos(\alpha) = \cos(\alpha_0) + \frac{-\operatorname{sen}(\alpha_0)}{1!}(\alpha-\alpha_0) + O(\alpha-\alpha_0)^2$$



$\alpha \rightarrow 0$



$$\cos(\alpha) = \overbrace{\cos(0)}^{1} + \overbrace{\left(\frac{-\operatorname{sen}(0)}{1}\right)}^{0}(\alpha-0) + \overbrace{O(\alpha\text{-}0)}^{\simeq 0}$$

$$\cos\alpha \simeq 1$$

$$\operatorname{sen}(\alpha) = \operatorname{sen}(\alpha_0) + \frac{\cos(\alpha_0)}{1!}\cdot(\alpha-\alpha_0) + O(\alpha-\alpha_0)$$

$$\operatorname{sen}(\alpha) = \overbrace{\operatorname{sen}(0)}^{0} + \overbrace{\frac{\cos(0)}{1!}}^{1}(\alpha-0) + \overbrace{O(\alpha-0)}^{\simeq 0}$$

$$\operatorname{sen}(\alpha) \simeq \alpha$$

$\begin{cases} \cos\theta = 1 \\ \cos\psi = 1 \end{cases}$ $\quad \operatorname{sen}\theta = \theta$, $\operatorname{sen}\psi = \psi$

$$\boldsymbol{\beta}_2 \cdot \boldsymbol{\beta}_1 = \begin{bmatrix} 1 & \psi & \theta \\ -\psi & 1 & -\theta\,\psi \\ -\theta & 0 & 1 \end{bmatrix} \qquad \boldsymbol{\beta}_1 \cdot \boldsymbol{\beta}_2 = \begin{bmatrix} 1 & \psi & \theta \\ -\psi & 1 & 0 \\ -\theta & -\theta\,\psi & 1 \end{bmatrix}$$

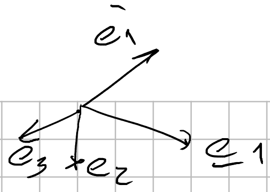
$\theta \rightarrow 0$
$\psi \rightarrow 0$

$\theta \cdot \psi \longrightarrow$ infinitesimal de orden mayor

$$\boldsymbol{\beta}_2 \cdot \boldsymbol{\beta}_1 = \begin{bmatrix} 1 & \psi & \theta \\ -\psi & 1 & 0 \\ -\theta & 0 & 1 \end{bmatrix} \qquad \boldsymbol{\beta}_1 \cdot \boldsymbol{\beta}_2 = \begin{bmatrix} 1 & \psi & \theta \\ -\psi & 1 & 0 \\ -\theta & 0 & 1 \end{bmatrix}$$

$\Downarrow$

Las pequeñas rotaciones, éstas son conmutativas



$$e'_1 \to \{\underline{e}_j)$$

Pongo las componentes de los versores e'_1 en la base $\underline{e}_j$.

$$B_{ij} = \underline{e}'_i \cdot \underline{e}_j$$

$$B_{ij} = (\underline{e}'_i)_j$$

$$\underline{\underline{B}} = \begin{bmatrix} [\ e'_1\ ] \\ [\ e'_2\ ] \\ [\ e'_3\ ] \end{bmatrix}$$